



MODULE 5



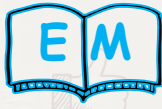


MODULE 5



SYLLABUS

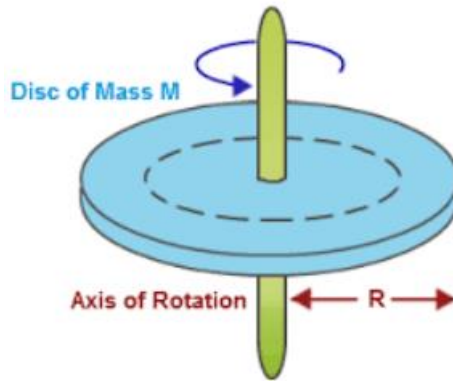
- ✓ Rotation – kinematics of rotation- equation of motion for a rigid body rotating about a fixed axis – rotation under a constant moment.
- ✓ Plane motion of rigid body – instantaneous centre of rotation (concept only).
- ✓ Simple harmonic motion – free vibration –degree of freedom- undamped free vibration of spring mass system- effect of damping(concept only)



ROTATION (CIRCULAR MOTION)

- ✓ When a body moves along a circular path, then it is called rotary motion.

Rotational Motion Examples



Top



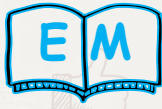
Soccer ball



Ferris wheel



Ice skater



CURVILINEAR MOTION & CIRCULAR MOTION

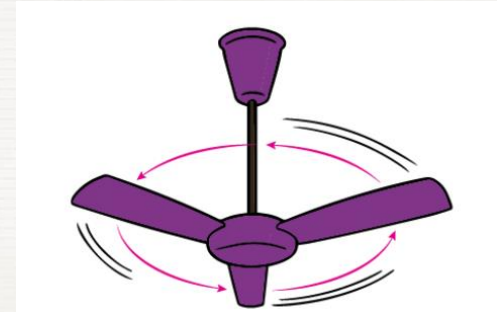
Curvilinear Motion

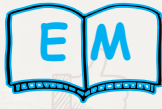
- ✓ The body does not repeat the same path again and again.
- ✓ The centre of rotation goes on changing along its path.



Circular Motion

- ✓ The body continues to repeat the same circular path.
- ✓ The centre of rotation is fixed.



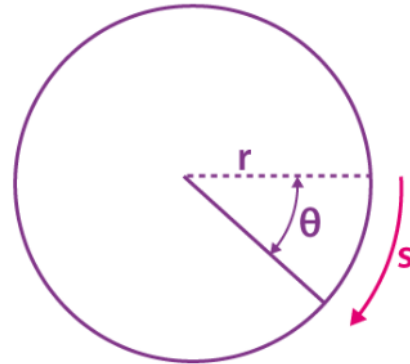


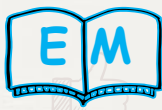
CIRCULAR MOTION

Angular Displacement (θ):

- ✓ Displacement of a particle in circular motion.
- ✓ It is the total angle (in radians) through which the particle has rotated.

Angular displacement





CIRCULAR MOTION

Angular Velocity (ω):

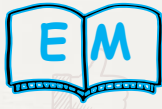
- ✓ Rate of change of angular displacement.
- ✓ Unit - rad/sec.

$$\text{Angular velocity} = \frac{\text{Angular displacement}}{\text{time}}$$

$$\text{Mathematically, } \omega = \frac{d\theta}{dt}$$

Angular velocity is also expressed in terms of rpm.

$$\text{ie, } \omega = N \text{ rpm} = \frac{2\pi N}{60} \text{ rad/s}$$



CIRCULAR MOTION

Angular Acceleration (α):

- ✓ Rate rate of change of angular velocity.
- ✓ May uniform or variable.

Angular Acceleration = $\frac{\text{Change of angular velocity}}{\text{Time}}$

$$\text{Mathematically, } \alpha = \frac{d\omega}{dt} = \frac{d}{dt} \left(\frac{d\theta}{dt} \right) = \frac{d^2\theta}{dt^2}$$

$$\text{or, } \alpha = \frac{d\omega}{dt} \times \frac{d\theta}{d\theta} = \frac{d\omega}{d\theta} \times \frac{d\theta}{dt} = \frac{d\omega}{d\theta} \times \omega$$

$$\text{ie, } \alpha = \omega \cdot \frac{d\omega}{d\theta}$$

Equations of Circular Motion of a body moving under constant Angular Acceleration

The equations relate angular displacement (θ), time taken (t), initial angular velocity (ω_0), final angular velocity (ω) and uniform angular acceleration (α).



Equations of
circular
motion

$$\omega = \omega_0 + \alpha t$$

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$



Relationship between Linear Motion & Circular Motion

Linear/Tangential Velocity & Angular Velocity:

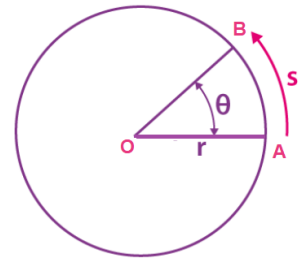
Consider a particle moving along a circular path. Let the particle be initially at 'A' and after time 't', let the particle be at 'B'.

Let $\angle AOB = \theta$

r - radius of circular path

ω - angular velocity

v - linear velocity (*tangential to circular path*)



Linear/Tangential Velocity & Angular Velocity:

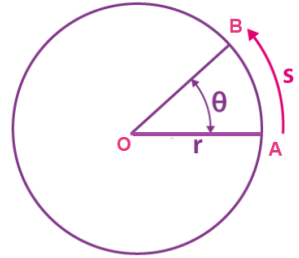
Linear velocity = $\frac{\text{Linear displacement}}{\text{time}}$

$$v = \frac{\text{arc AB}}{t}$$

$$= \frac{r\theta}{t}$$

$$= r\omega$$

$$; S = r.\theta$$



ie, $v = r.\omega$

Tangential Acceleration & Angular Acceleration:

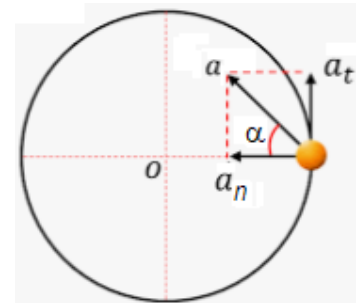
We have, $v = r \omega$

Differentiating w.r.t time, $\frac{dv}{dt} = r \cdot \frac{d\omega}{dt}$

ie, $a_t = r\alpha$

Normal/ Radial/ Centripetal Acceleration:

$$\begin{aligned} a_n &= v \cdot \omega \\ a_n &= r \cdot \omega^2 \\ a_n &= \frac{v^2}{r} \end{aligned}$$



Total Acceleration:

We have, a_t - tangential acceleration

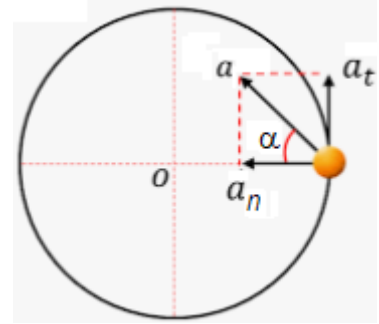
a_n - normal acceleration

∴ Total acceleration,

$$a = \sqrt{a_t^2 + a_n^2}$$

$$\tan \alpha = \frac{a_t}{a_n}$$

$$\therefore \alpha = \tan^{-1} \left(\frac{a_t}{a_n} \right)$$





D'Alembert's Principle in Circular Motion

D'Alembert's Principle states that "When the external torque acts on a system causing circular motion, the algebraic sum of all the torques acting on the system due to external forces and the inertia torque is zero".

$$\text{ie, } T + (-I \alpha) = 0.$$

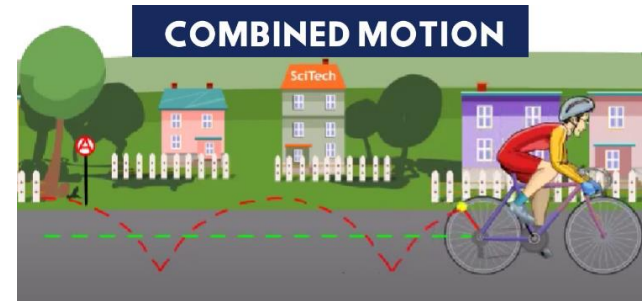
This equation is called dynamic equation of equilibrium.

TYPES OF MOTION

A body may have the following types of motion:

- Motion of translation
- Motion of rotation
- Combined motion of translation and rotation

(General plane motion)



i) Motion of translation or plane motion

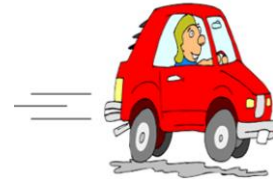
- All its particles move in parallel path and travels the same distance.
- All the particles of the body have same velocity and same acceleration at any particular instant.

The translational motion can be;

a. Rectilinear

b. Curvilinear

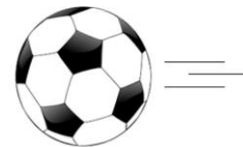
Translational Motion Examples



Car



Apple



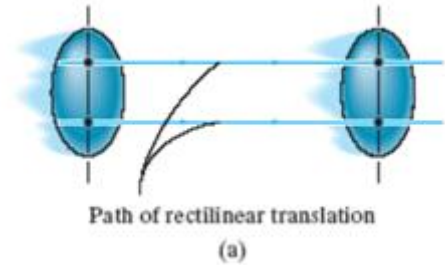
Soccer ball



Train

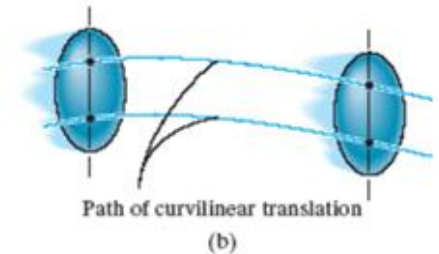
a. Rectilinear translation

All the particles move in straight parallel path.



b. Curvilinear translation

All the particles move in curved parallel path.



ii) Motion of rotation

- Body rotates about a fixed point in such a way that all its particles move in circular path.
- The fixed point about which the body rotates is called center of rotation and the axis passing through the fixed point is known as axis of rotation.
- The particles lying on the axis of rotation have zero velocity and zero acceleration.

Rotational Motion Examples



Top



Soccer ball



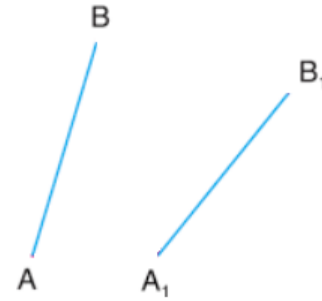
Ferris wheel



Ice skater

iii) Combined motion of translation and rotation

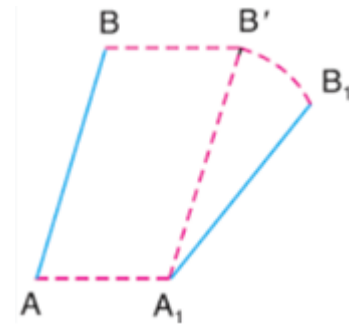
- Consider a link AB which moves from its initial position AB to final position A_1B_1 in a short interval of time as shown.



- The link has neither entire motion of translation nor entire motion of rotation, but a combination of the two.

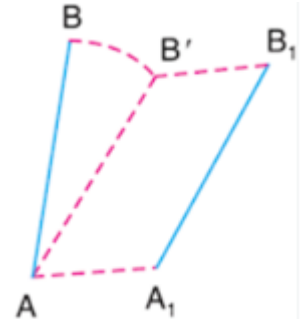
iii) Combined motion of translation and rotation

- If we split up the motion of link AB, we will find that the link have first motion of translation from AB to A_1B' and then motion of rotation about A_1 till it occupies the final position A_1B_1 as shown in figure.



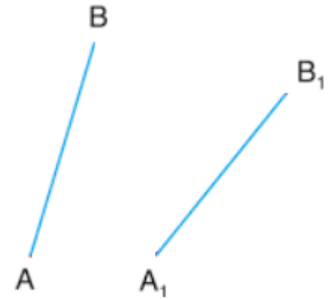
iii) Combined motion of translation and rotation

- The motion of the link AB may also be considered to have first the motion of rotation from AB to AB' to A_1B_1 as shown in the fig.
- Such a motion of the link from AB to A_1B_1 is an example of combined motion of rotation and translation, it being immaterial whether the motion of rotation takes place first or the motion of translation.



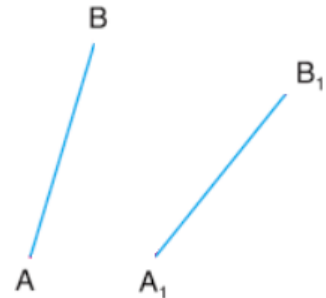
Instantaneous Centre of Rotation

- The combined motion of link AB from its initial position AB to the position A_1B_1 may be assumed to be a motion of rotation about some centre.
- As the position of the link AB goes on changing, the centre about which the motion of rotation is assumed to take place also goes on changing. Such a centre which goes on changing from instant to instant is known as instantaneous centre.



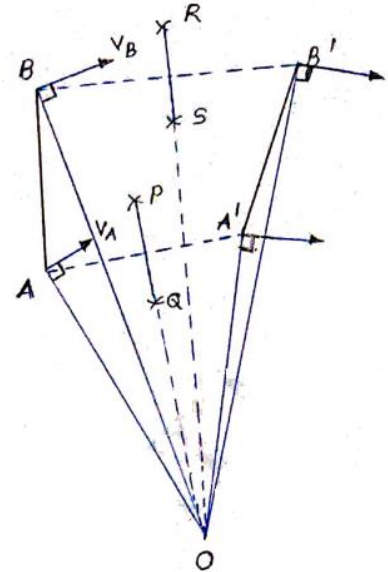
Instantaneous Centre of Rotation

- The instantaneous centre of rotation is instantaneously at rest and has zero velocity.
- Hence this point is also known as instantaneous centre of zero velocity.
- The link may seem to be rotating about one point at one instant of time and about another point at the next instant.
- The locus of the instantaneous centre as the link goes on changing its position is called centrode.



Position of the instantaneous centre (Graphical Method)

- i) Draw initial and final position of the rigid link AB and A'B' respectively.
- ii) Join AA' and BB'.
- iii) Draw PQ - perpendicular bisector of AA' and RS - perpendicular bisector of BB'.
- iv) Extend the two particular bisector PQ and RS so as to meet at point O, which is the required centre of rotation or instantaneous centre.



Instantaneous centre of rotation

Let ω - angular velocity of the link AB about O

V_A - Linear velocity of point A

V_B - Linear velocity of point B

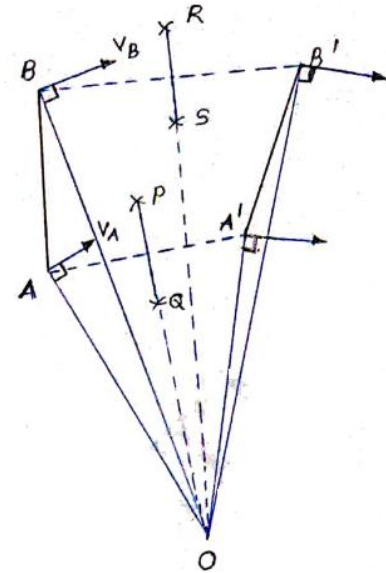
V_A = distance of point A from the centre of rotation $\times$ angular velocity

$$\text{ie, } V_A = OA \times \omega \quad \dots\dots\dots (i)$$

$$\text{Also, } V_B = OB \times \omega \quad \dots\dots\dots (ii)$$

$$\frac{(i)}{(ii)} \rightarrow \frac{V_A}{V_B} = \frac{OA}{OB}$$

If directions of V_A and V_B are known, then instantaneous centre of AB is obtained by drawing perpendiculars to the directions of V_A and V_B .





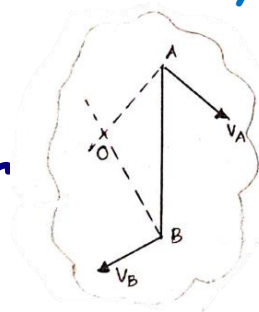
Properties of Instantaneous centre

- The velocity of a point on the body is proportional to the distance of the point from the instantaneous centre and the angular velocity of the body.
- When the direction of the linear velocities of two points of a body is known, then the location of the instantaneous centre can be found out by drawing perpendicular to the direction of linear velocities.

Location of the instantaneous centre for some general cases

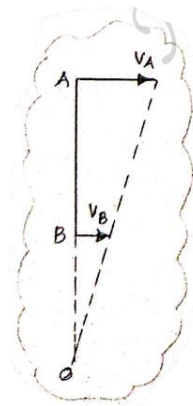
Case 1: When the direction of velocities of any two points on the body are known.

- Point of intersection of the lines drawn perpendicular to the direction of velocities.



Case 2: When the direction of velocities are parallel and unequal but in same direction.

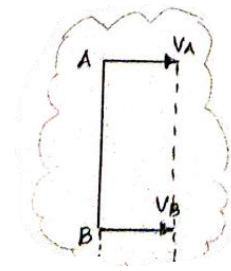
- Point of intersection of lines joining the tip of the velocity vectors and the extension of the line joining the points.



Location of the instantaneous centre for some general cases

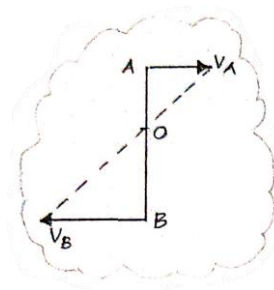
**Motion of translation:*

(direction of velocities parallel, equal in magnitudes,
same direction)



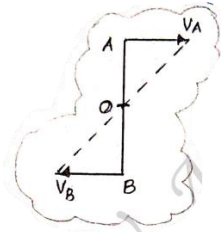
Case 3: When the direction of velocities are parallel, equal and opposite.

- Point of intersection of the line joining the tip of velocity vectors and the line joining the points.



**Motion of pure rotation:*

(direction of velocities are parallel, equal but in opposite
direction)





Characteristics of Instantaneous centre

- The instantaneous centre maybe located inside or outside of the body.
- The location of the instantaneous centre changes at every constant and path traced by it is called centrode.
- The instantaneous centre is having no velocity.
- It is used only for velocity calculation.



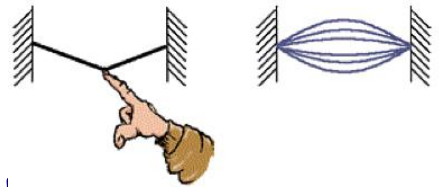
Mechanical Vibrations

- Motion of particle or a body which oscillates about a position of equilibrium.
- It results when a system is displaced from a position of a stable equilibrium.
- The system tends to return of this position under the action of restoring force.

Mechanical system may undergo **free vibrations** or they may be subjected to the **forced vibrations**.

Free Vibrations

- If a disturbing force is applied to a system to start the vibration and then removed from the system leaving it to vibrate by itself, then the system is said to undergo **free vibration or natural vibration**. eg. Oscillation of simple pendulum.
- The frequency of this vibration is known as the **natural frequency** of the system.





Forced Vibrations

- If the disturbing force continues to act at periodic intervals on the system, the system is said to undergo **forced vibration**.

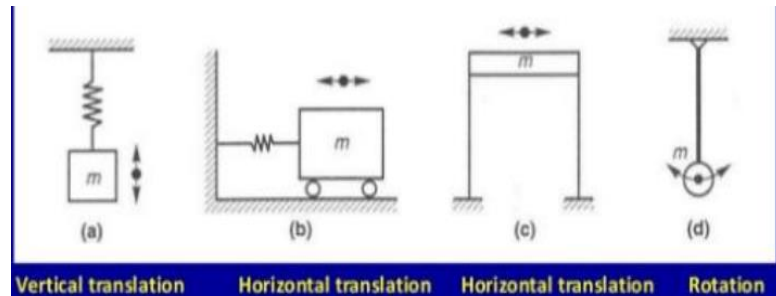
eg. A child's swing that is pushed on each downswing.

Degrees of freedom (DOF)

- It is the number of independent coordinates to describe the motion of a mass or a system.

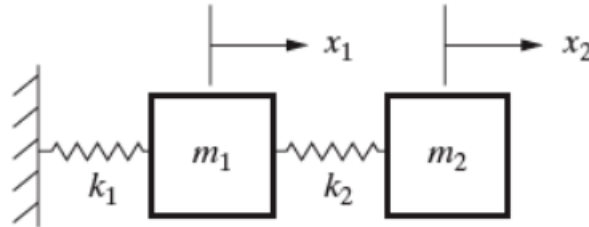
Single degree of freedom:

- A mass or system which requires only one independent coordinate to describe its motion.



Multiple degree of freedom:

- A mass or system which requires two or more independent coordinates to describe its motion.

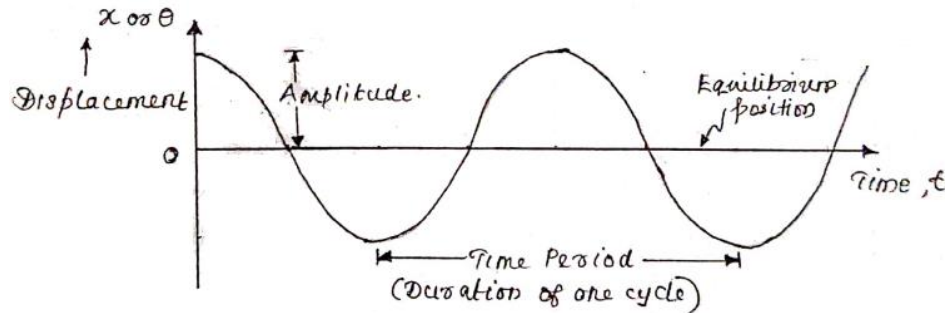


Periodic motion/ Vibratory motion/ Oscillatory motion

- A motion which repeats itself in equal intervals of time is called periodic motion.

eg. Rise and fall of tides, oscillation of electric circuit, vibrations of strings of a piano.

Displacement-time plot:

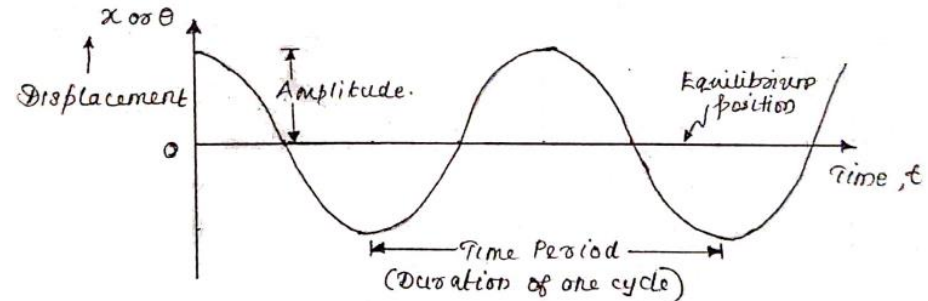


Time period

- Time interval required to execute one cycle (one complete to and fro motion). Also the time required to describe an angle of 2π radians.

One cycle

- One complete repetition of the motion is known as one cycle.





Frequency or Cyclic frequency (f or N or v)

It is the number of cycles executed per unit time and is the inverse of time period.

$$f \text{ or } N = \frac{1}{T}$$

Unit - cycles/sec or Hertz (Hz)

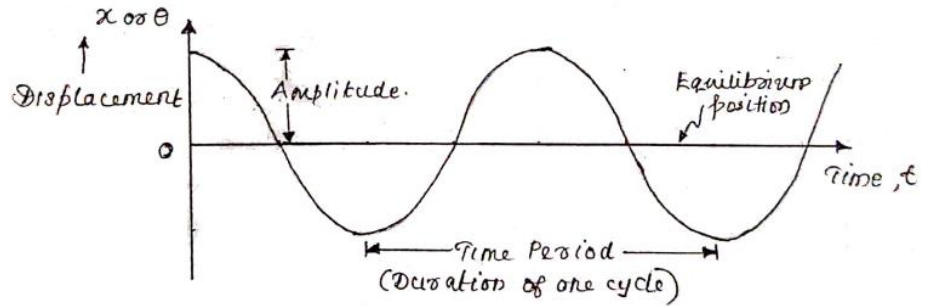
Angular or Circular frequency (ω)

Angular frequency is 2π times cyclic frequency.

$$\omega = 2\pi N = 2\pi \frac{1}{T}$$

Amplitude

Amplitude of a periodic motion is the maximum displacement of the system from its equilibrium position.





SIMPLE HARMONIC MOTION (SHM)

- Simplest type of periodic motion.
- The particles move to and fro, either along a straight line or along the arc of a circle, about a fixed point called the equilibrium or mean position.
- When the vibrating body of a system has to and fro motion along a straight line, the vibration is called linear vibration or longitudinal vibration.



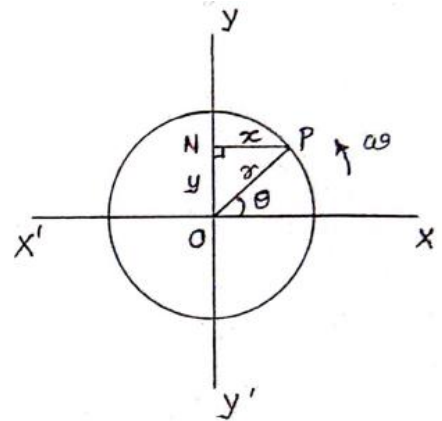
Characteristics of SHM

- 1) The motion is oscillatory about a point lying on the path of motion is called mean position.
- 2) The acceleration of the particle is always proportional to the distance of the particle from the mean position.
- 3) The acceleration of a particle executing SHM is always directed towards the mean position.

Velocity and Acceleration of a particle moving with SHM

SHM may be defined as the projection of a uniform circular motion on any diameter of the circle.

Consider a particle moving along the circumference of a circle of radius ' r ', with a uniform angular velocity of ' ω ' rad/s as shown in the figure.



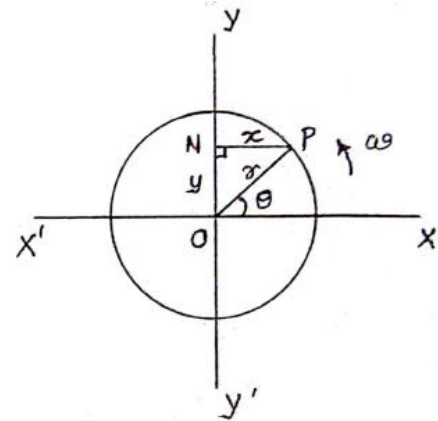
Velocity and Acceleration of a particle moving with SHM

Let 'P' be the position of the particle at some instant after 't' sec from x.

∴ Angle turned by the particle,
 $\theta = \omega t$ radians

Displacement of point N,
 $y = ON = r \sin\theta$

(ie, projection of 'P' on the vertical diameter y-y' of circle)
where, r - amplitude of SHM



Velocity and Acceleration of a particle moving with SHM

$$y = r \sin \theta$$

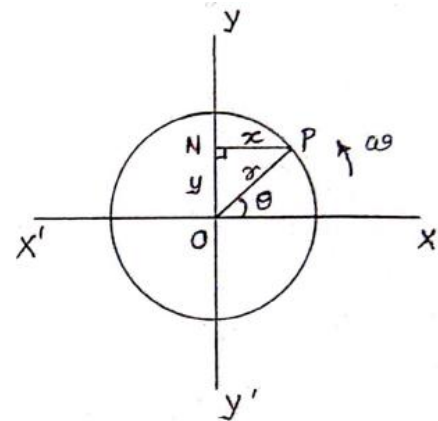
$$\text{ie, } y = r \sin \omega t \text{ -----(1)}$$

Differentiating eqn (1) w.r.t time,

$$\frac{dy}{dt} = r\omega \cos \omega t$$

$$\text{ie, velocity, } v_y = r\omega \cos \omega t \text{ -----(2)}$$

$$\text{ie, } v_y = r\omega \sqrt{1 - \sin^2 \omega t}$$



Velocity and Acceleration of a particle moving with SHM

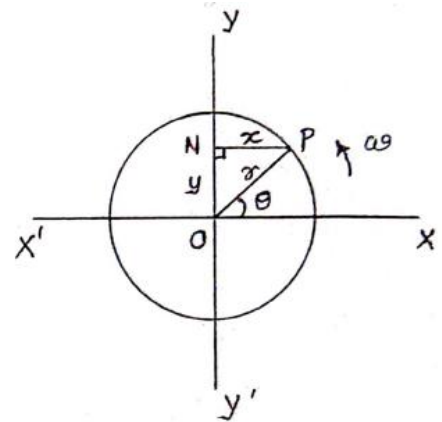
$$\text{But (1)} \rightarrow \sin \omega t = \frac{y}{r}$$

$$\text{ie, } v_y = r\omega \sqrt{1 - \sin^2 \omega t}$$

$$= r\omega \sqrt{1 - \frac{y^2}{r^2}}$$

$$\text{ie, } v_y = \omega \sqrt{r^2 - y^2}$$

$$y = r \sin \omega t \text{ ---(1)}$$





Velocity and Acceleration of a particle moving with SHM

Differentiating eqn (2) w.r.t time,

$$v_y = r\omega \cos \omega t \text{ --(2)}$$

$$\frac{dv_y}{dt} = \frac{dy^2}{dt^2} = -r\omega^2 \sin \omega t$$

$$\therefore a_y = -\omega^2 y$$

; where $y = r \sin \omega t$

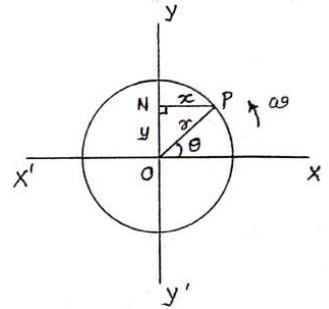
Similarly, $v_x = \omega \sqrt{r^2 - x^2}$

ie, $a_x = -\omega^2 x$

Velocity and Acceleration of a particle moving with SHM

NOTE:

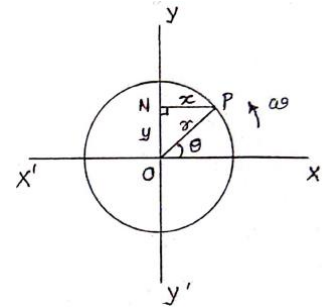
- The minus sign shows that the direction of acceleration is opposite to the direction in which y increases. ($a_y = -\omega^2 y$)
- ie, the acceleration is always directed towards the point O . But in the actual practice, this relation is used as $a = \omega^2 y$.



Maximum and Minimum velocity & Acceleration of a particle moving with SHM.

Velocity of a particle moving with SHM,

$$v = \omega \sqrt{r^2 - y^2}$$



Velocity is maximum, when $y = 0$ or when N passes through 'O', ie, it's mean position.

Max Velocity, $V_{\max} = r\omega$

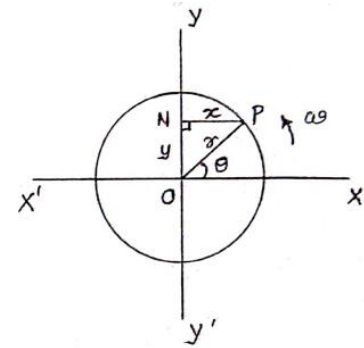
Velocity is zero, when $y = r$, ie when N passes through y or y' .

Maximum and Minimum velocity & Acceleration of a particle moving with SHM.

Acceleration of SHM, $a = \omega^2 y$

Acceleration is maximum, when $y = r$. ie, when N passes through y' or y .

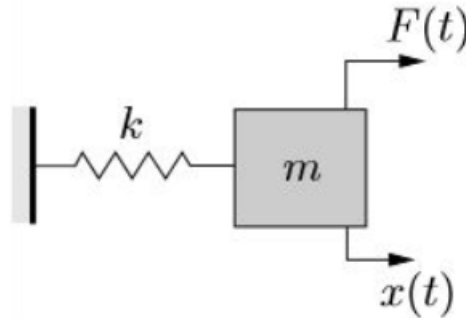
ie, $a_{\max} = \omega^2 r$



Acceleration is zero, when $y = 0$, ie, when N passes through O, i.e., mean position.

SPRING MASS MODEL

A spring mass combination with the mass subjected to a force along the axis of the spring and passing through the centre of mass of the spring is called a linear spring mass system.



Natural frequency of a free linear vibration

Consider a spring of stiffness k , when a body of mass ' m ' is attached to one end of the spring with the other end and fixed, the spring elongates.

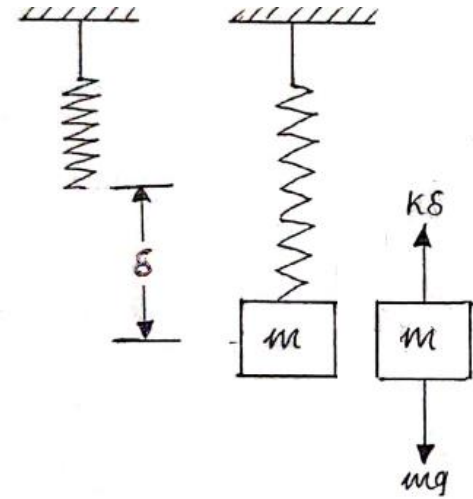
Let δ - the elongation of the spring.

(static deflection of the spring)

In the equilibrium position, net force on the body is zero.

$$\text{ie, } mg - k\delta = 0$$

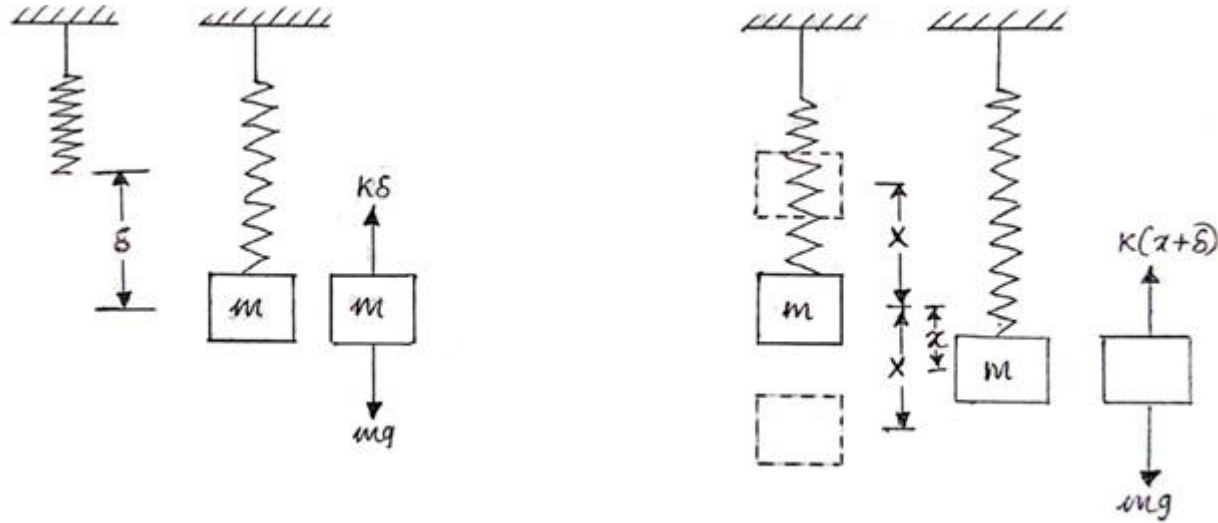
$$mg = k\delta$$



k - Stiffness of spring or Force required to produce unit deflection.

$k\delta$ - Spring force opposite to the direction of displacement.

Natural frequency of a free linear vibration



When the body is displaced from the equilibrium position by an external force and if the external force is removed, then the body will vibrate between two extreme position with amplitude 'X'.

Natural frequency of a free linear vibration

Consider the position of the body when it is at a distance 'x' below the equilibrium position. As the body is in motion net force is equal to 'mass x acceleration'.

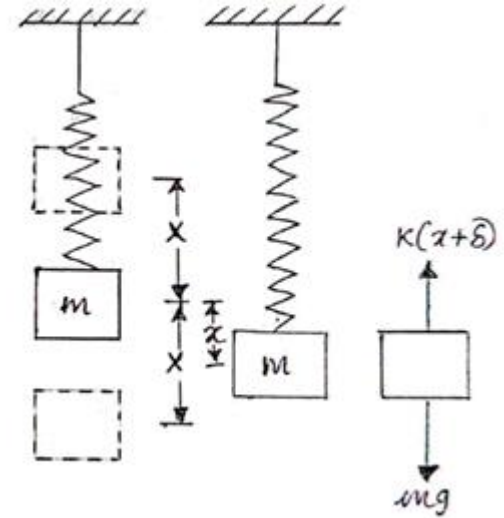
$$\text{ie, } mg - k(x + \delta) = m.a$$

$$mg - kx - k\delta = m.a$$

$$k\delta - kx - k\delta = m.a$$

$$-kx = m.a$$

$$\text{ie, } a = -\frac{k}{m}.x$$



$$[\because mg = k\delta]$$



Natural frequency of a free linear vibration

$$a = -\frac{k}{m} \cdot x \quad \dots\dots\dots(1)$$

ie, $a + \frac{k}{m} \cdot x = 0$

This is the equation of motion of a vibrating body.

Comparing eqn (1) with the equation of motion of SHM, $a = -\omega^2 x \quad \dots\dots\dots(2)$

$$(1) \ \& \ (2) \rightarrow \omega^2 = \frac{k}{m} ;$$



Natural frequency of a free linear vibration

$$\omega^2 = \frac{k}{m} \rightarrow \boxed{\omega = \sqrt{\frac{k}{m}}}$$

As the system vibrates freely, this frequency is called **natural frequency**, ω_n .

$$\omega_n = \sqrt{\frac{k}{m}} \rightarrow f_n = \frac{\omega_n}{2\pi} \rightarrow \boxed{f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}}}$$

This is the expression for **natural cyclic frequency** of free linear vibration.

Springs connected in Series and Parallel

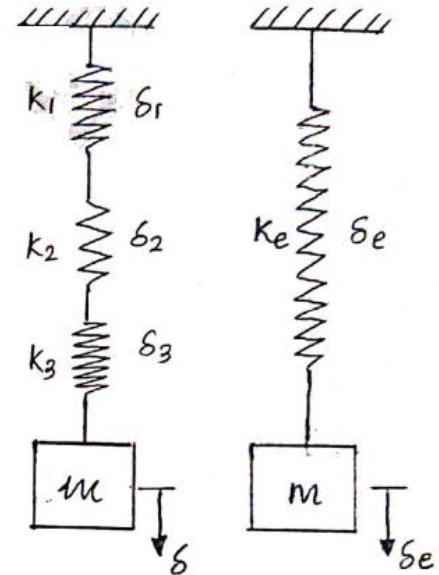
In a spring mass model, the springs can be attached to the mass either in series or in parallel.

Springs in Series

A number of spring having stiffness k_1, k_2, k_3 etc. arranged in series can be replaced by a single spring of stiffness,

' k_e ' - equivalent stiffness. ($\delta_e = \delta$)

Let $\delta_1, \delta_2, \delta_3$ be the elongation of each spring due to body of mass m .



Springs in Series

Static deflection of mass, $\delta = \delta_1 + \delta_2 + \delta_3$

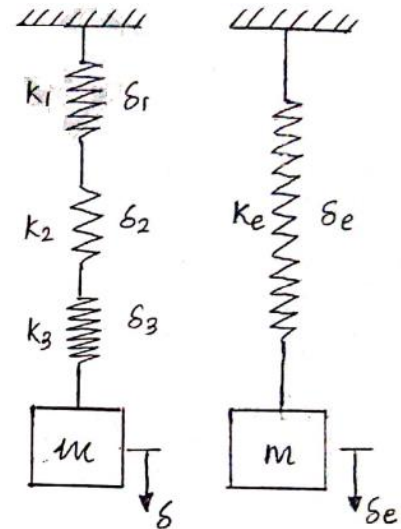
Let δ_e - static deflection of same mass when attached to a single spring of stiffness - k_e .

$$\text{ie, } \delta_e = \delta = \delta_1 + \delta_2 + \delta_3 \dots (1)$$

$$\text{But } mg = k \delta$$

$$\therefore \delta_e = \frac{mg}{k_e} ; \delta_1 = \frac{mg}{k_1} ; \delta_2 = \frac{mg}{k_2} ; \delta_3 = \frac{mg}{k_3}$$

$$(1) \rightarrow \boxed{\frac{1}{k_e} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3}}$$



Springs in parallel

Fig. shows three springs arranged in parallel. The reqd condition for the single spring to be equivalent to the springs connected in parallel is $\delta_e = \delta$.

For equilibrium, $mg = k_1\delta + k_2\delta + k_3\delta$

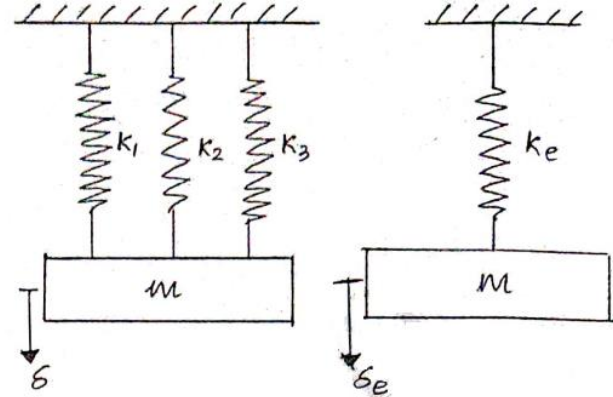
$$\text{ie, } mg = (k_1 + k_2 + k_3)\delta \dots\dots\dots(1)$$

$$\text{Also, } mg = k_e \delta_e \dots\dots\dots(2)$$

$$(1) \ \& \ (2) \rightarrow k_e \delta_e = (k_1 + k_2 + k_3)\delta$$

$$\text{As } \delta_e = \delta, \quad \boxed{k_e = k_1 + k_2 + k_3}$$

where, k_e - equivalent stiffness.





Damped and undamped free vibrations

Undamped Vibration

- No resistive force acts on the vibrating object.
- Energy in the object is continuously transformed from kinetic energy to potential energy.
- Sum of kinetic and potential energy remains a constant value.

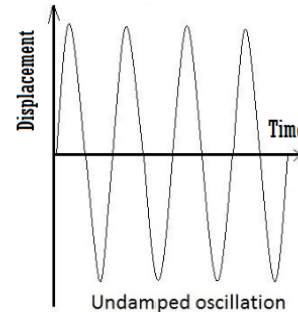
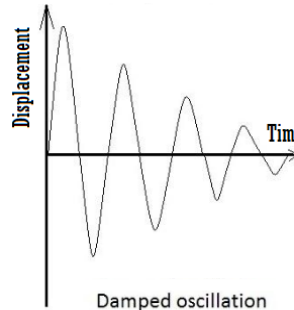
In practice, it's extremely difficult to find undamped vibrations. For instance, even an object vibrating in air would lose energy over time due to air resistance.

Damped and undamped free vibrations

Damped Vibration

- External resistive forces act on the vibrating object.
- The object loses energy due to resistance and as a result, the amplitude of vibrations decreases exponentially.

Displacement - Time curve for damped & undamped oscillation





Damped and undamped free vibrations

Damped Vibration	Undamped Vibration
Vibrating object loses its energy to the surroundings.	Energy of the vibrating object does not get dissipated to surroundings over time.